

INTRODUCTION TO EMBEDDED SYSTEMS & IoT DEVICE DESIGN

Embedded Systems Task-1

Project Report on

RESEARCH ON EMBEDDED SYSTEMS CONCEPTS AND SMART TEMPERATURE MONITORING SYSTEM

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Project Domain:

Embedded Systems & Internet of Things (IoT)

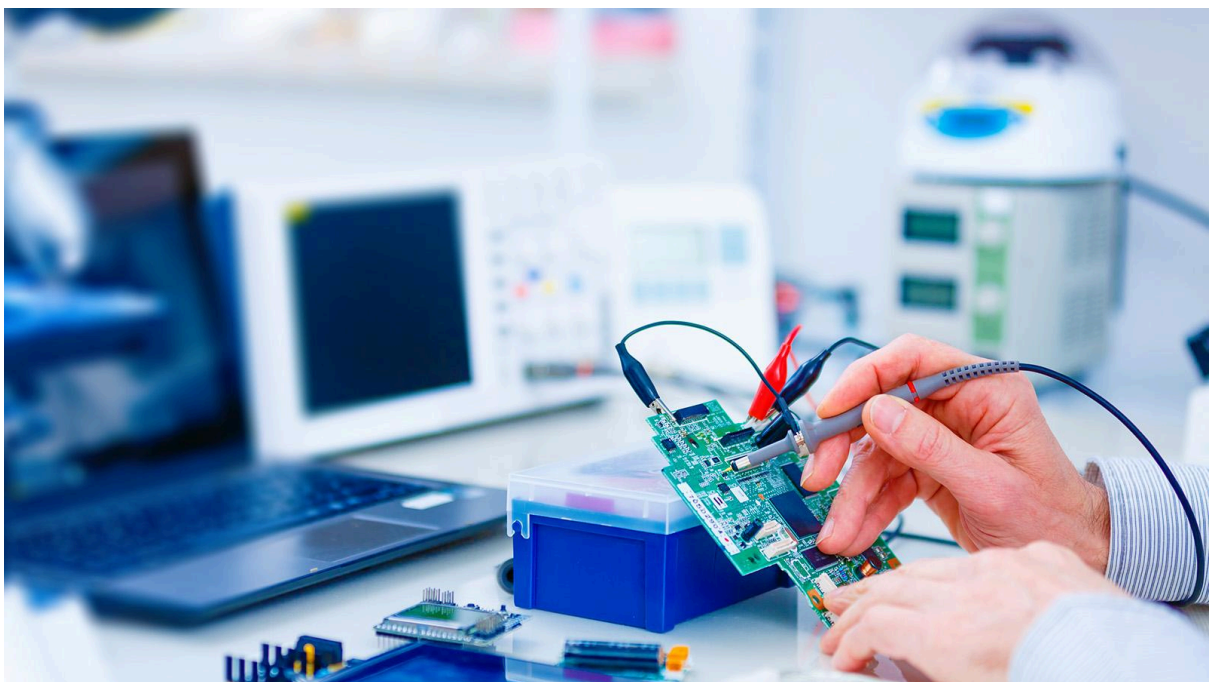
1. INTRODUCTION SECTION (RESEARCH)

What is an Embedded System?

An embedded system is a specially designed computer system developed to perform a specific task or a set of dedicated tasks within a larger electronic or mechanical system. Unlike general-purpose computers such as desktops or laptops that can run multiple applications for different purposes, embedded systems are built for particular operations such as monitoring, controlling, sensing, processing, or automation.

An embedded system combines both hardware and software components into a single integrated unit. The hardware part generally includes a microcontroller or microprocessor, memory units, sensors, actuators, input/output interfaces, and communication modules. The software part, often called firmware, contains the instructions required to control the hardware and perform the intended operation.

Embedded systems are designed to operate efficiently with minimal human intervention. They are optimized for high reliability, fast performance, low power consumption, and compact size. These systems are capable of continuously monitoring inputs from sensors, processing the data, and generating suitable outputs in real time.



The growing advancement of technology has made embedded systems an important part of modern life. Today, almost every smart electronic device uses embedded systems for intelligent operation and automation. These systems are used in consumer electronics, industrial automation, medical devices, automobiles, aerospace systems, telecommunications, and robotics.

Embedded systems provide several advantages such as reduced hardware size, lower manufacturing cost, better performance, and improved energy efficiency. Due to these features, they are widely adopted in portable and battery-powered devices.

Examples of embedded systems include washing machines, microwave ovens, traffic light controllers, digital cameras, smart televisions, fitness bands, printers, ATM machines, industrial robots, and smart home devices.

Real-World Applications of Embedded Systems

Smart Home Systems

Embedded systems play an important role in smart home automation. Modern homes use intelligent devices that can automatically monitor and control household operations. Smart lighting systems can switch lights ON or OFF automatically depending on motion or ambient light conditions. Smart thermostats regulate room temperature efficiently to reduce energy consumption. Smart door locks provide secure access control using passwords, RFID, or mobile applications. Home security systems use embedded controllers with sensors and cameras for surveillance and intrusion detection.

Automotive Industry

The automotive industry heavily depends on embedded systems for vehicle safety, performance, and automation. Modern vehicles contain multiple embedded controllers that continuously monitor and control different vehicle operations.

Embedded systems are used in engine control units (ECUs) for fuel injection and engine performance optimization. Anti-lock braking systems (ABS) prevent wheel locking during sudden braking and improve vehicle safety. Airbag control systems detect collisions and deploy airbags instantly to protect passengers. Parking assistance systems use ultrasonic sensors and cameras for obstacle detection.

Healthcare and Medical Devices

Medical equipment uses embedded systems for accurate monitoring, diagnosis, and treatment. These systems help healthcare professionals provide better patient care and improve medical safety.

Devices such as heart rate monitors, ECG machines, glucose monitoring systems, ventilators, infusion pumps, and digital thermometers use embedded systems for data processing and control. Wearable fitness devices continuously monitor health parameters such as pulse rate, oxygen levels, and body temperature.

Embedded systems in medical devices must operate with high accuracy, reliability, and real-time performance because human lives depend on their proper functioning.

Industrial Automation

Industries use embedded systems extensively for process automation, monitoring, and machine control. Industrial embedded systems increase productivity, reduce human effort, and improve operational efficiency.

These systems are used in conveyor belt automation, robotic assembly systems, temperature control systems, pressure monitoring systems, and machine safety mechanisms. Embedded controllers continuously monitor industrial parameters using sensors and take necessary actions automatically.

Programmable logic controllers (PLCs), industrial robots, and automated production systems are examples of embedded systems used in manufacturing industries.

Robotics

Robotics is one of the most important applications of embedded systems. Robots use embedded controllers for processing sensor data, controlling motors, navigation, and decision-making.

Embedded systems help robots detect obstacles, follow paths, maintain balance, and interact with the environment. Autonomous robots use sensors such as ultrasonic sensors, cameras, and infrared sensors to collect environmental data and process it in real time.

Robots are widely used in industries, healthcare, defense, agriculture, and space research for performing complex or dangerous tasks.

Consumer Electronics

Consumer electronic devices use embedded systems for smart and efficient operation. Devices such as smart televisions, washing machines, refrigerators, microwave ovens, air conditioners, gaming consoles, and digital cameras contain embedded controllers.

These systems automate device functions, improve user interaction, and provide advanced features while consuming less power.

Telecommunications

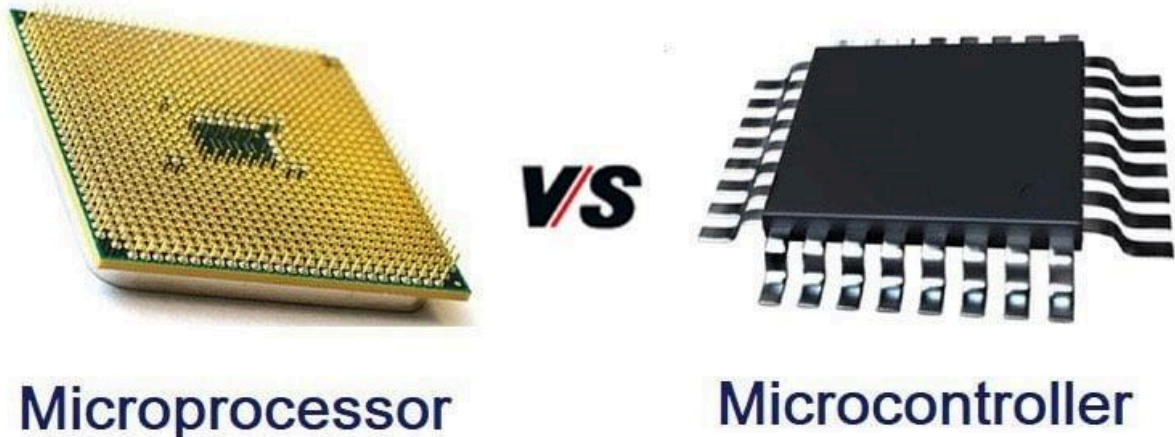
Embedded systems are essential in communication devices and networking equipment. Routers, switches, modems, smartphones, and satellite communication systems use embedded processors for signal processing and data communication.

These systems enable fast and reliable communication over wired and wireless networks.



Difference Between Microcontroller and Microprocessor

A microcontroller and a microprocessor are both electronic computing devices used in digital systems, but they differ significantly in architecture, functionality, and applications.



A microcontroller is a compact integrated circuit that contains a processor, memory, and input/output peripherals on a single chip. It is mainly designed for embedded applications where dedicated control operations are required. Microcontrollers are widely used in automation systems, sensor-based applications, and portable electronic devices because they consume less power and require minimal external components.

A microprocessor, on the other hand, mainly consists of a central processing unit (CPU). It requires external memory, input/output interfaces, and supporting peripherals for operation. Microprocessors are designed for high-performance general-purpose computing systems such as computers and laptops.

Microcontrollers are cost-effective, energy-efficient, and suitable for real-time control systems, whereas microprocessors provide higher computational power and are used in complex applications requiring multitasking and advanced processing capabilities.

Examples of microcontrollers include Arduino UNO, ESP32, PIC, and AVR controllers, while examples of microprocessors include Intel Core i5, Intel i7, and AMD Ryzen processors.

2. Component Study

Research and explaining the role of Microcontrollers (Arduino, ESP32, Raspberry Pi Pico).

Arduino UNO:

Arduino UNO is one of the most popular microcontroller development boards used for learning embedded systems and IoT projects. It is based on the ATmega328P microcontroller and provides an easy-to-use platform for beginners and developers.



The board contains digital input/output pins, analog input pins, USB connectivity, power regulation circuits, and programming support through the Arduino IDE. Users can easily write and upload programs using C/C++ language.

Arduino UNO is commonly used in sensor interfacing, robotics, automation systems, home automation projects, and educational experiments. Its simplicity and large community support make it ideal for beginners in embedded systems.

ESP32:

ESP32 is an advanced microcontroller developed for IoT and wireless communication applications. It includes built-in Wi-Fi and Bluetooth connectivity, making it highly suitable for smart devices and internet-based systems.

ESP32 contains a powerful dual-core processor, multiple GPIO pins, analog and digital interfaces, PWM support, and low-power operating modes. It can connect directly to cloud platforms and mobile applications without requiring additional communication modules.



ESP32 is widely used in smart home systems, wireless sensor networks, industrial monitoring systems, remote automation, wearable devices, and IoT cloud applications.

RASPBERRY PI PICO:

The Raspberry Pi Pico is a low-cost microcontroller board developed using the RP2040 microcontroller chip. It is designed for embedded development, educational applications, and hardware prototyping.



The board provides multiple GPIO pins, programmable interfaces, analog inputs, and flexible communication support. It can be programmed using C/C++ and MicroPython languages. Raspberry Pi Pico is used in sensor-based projects, robotics, automation systems.

BASIC SENSORS

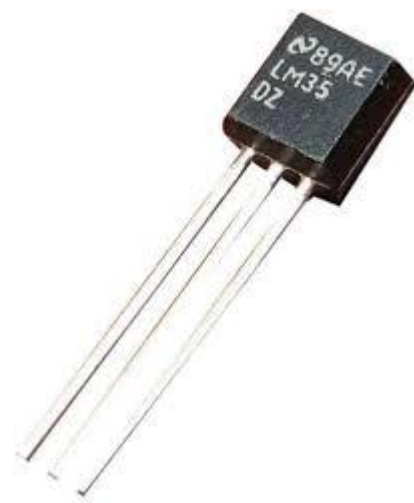
Sensors are electronic devices used to detect physical parameters from the environment and convert them into electrical signals that can be processed by a microcontroller.

Sensors are important components of embedded systems because they allow systems to interact with the real world.

Temperature Sensor:

A temperature sensor measures environmental or object temperature and converts it into an electrical signal. Temperature sensors are commonly used in weather monitoring systems, air conditioning systems, refrigerators, industrial process control, and healthcare equipment.

The LM35 temperature sensor is one of the most widely used analog temperature sensors. It provides output voltage proportional to temperature in degrees Celsius and can be easily interfaced with microcontrollers.



The LM35 is a high-precision, 3-pin analog temperature sensor integrated circuit manufactured by [Texas Instruments](https://www.ti.com). It provides a linear voltage output directly proportional to the Celsius (Centigrade) temperature with a scale factor of 10mV/C. This means at 25 C, the sensor outputs 250mV, making it incredibly simple to interface with microcontrollers without requiring complex calibration formulas.

Temperature sensors help embedded systems monitor thermal conditions and automatically control cooling or heating mechanisms.

Motion Sensor:

A motion sensor detects movement or motion in a particular area. These sensors are widely used in security systems, automatic doors, smart lighting systems, and surveillance systems.

The Passive Infrared (PIR) sensor is a commonly used motion sensor that detects infrared radiation emitted by human bodies or moving objects. When motion is detected, the sensor sends signals to the microcontroller, which can activate alarms, lights, or other devices.



HC-SR501 PIR MOTION SENSOR:

This is the most popular, versatile, and widely documented motion sensor module for DIY electronics.

- Operating Voltage: 5V to 20V DC
- Output Signal: High/Low digital signal (3.3V TTL output)
- Detection Range: Up to 7 meters (adjustable via potentiometer)
- Delay Time: 5 seconds to 5 minutes (adjustable via potentiometer)
- Trigger Modes: Has a jumper to select between Single Trigger (stops detecting after output goes High) and Repeat Trigger (constantly resets the delay timer as long as motion continues).

Motion sensors improve automation, energy efficiency, and security in embedded applications.

Light Sensor

A light sensor detects the intensity of light in the surrounding environment. It helps embedded systems automatically adjust operations based on ambient lighting conditions.



An LDR (Light Dependent Resistor) is a passive electronic component whose electrical resistance decreases as the intensity of surrounding light increases. Also known as a photoresistor, it is widely used in electronic circuits as a light sensor.

The Light Dependent Resistor (LDR) is a commonly used light sensor whose resistance changes according to light intensity. When light falls on the sensor, its resistance decreases, allowing the microcontroller to measure light levels.

Light sensors are widely used in automatic street lighting systems, brightness control systems, solar tracking systems, and smart energy-saving devices.

3. Mini Project – Smart Temperature Monitoring system

Project Title:

Smart Temperature Monitoring System Using TMP36 Sensor and Arduino UNO

Aim of the Project:

The aim of this project is to design and develop a smart temperature monitoring system using an Arduino UNO microcontroller and a TMP36 temperature sensor.

The system continuously monitors the surrounding temperature and provides an alert indication using an LED whenever the temperature exceeds a predefined threshold value.

This project demonstrates the implementation of embedded systems and sensor interfacing concepts using Tinkercad software in a virtual environment.

Components Required:

Component	Quantity
Arduino UNO	1
TMP36 Temperature Sensor	1
LED	1
330Ω Resistor	1
Breadboard	1
Jumper Wires	Required
USB Cable	1

Software Requirements:

- Arduino IDE.
- Tinkercad Circuits.

Working Principle:

The working principle of the Smart Temperature Monitoring System is based on temperature sensing, analog signal conversion, and threshold-based alert generation.

The TMP36 is a low-voltage analog temperature sensor that provides output voltage directly proportional to temperature in degrees Celsius. It is widely used in embedded and IoT applications because of its simplicity, accuracy, and low power consumption.

The sensor produces an analog voltage output where:

- 0.5V corresponds to 0°C
- Every 10mV increase represents a 1°C rise in temperature

The TMP36 sensor continuously detects the surrounding temperature and converts it into an analog voltage signal. This analog signal is sent to the Arduino UNO through analog pin A0.

The Arduino contains an internal Analog-to-Digital Converter (ADC) that converts the incoming analog voltage into a digital value. Since the Arduino UNO uses a 10-bit ADC, the sensor values range from 0 to 1023.

The Arduino program converts the ADC value into voltage using the reference voltage of 5V.

After calculating the voltage, the temperature is determined using the TMP36 temperature conversion formula.

The calculated temperature values in Celsius and Fahrenheit are displayed on the Serial Monitor for real-time monitoring.

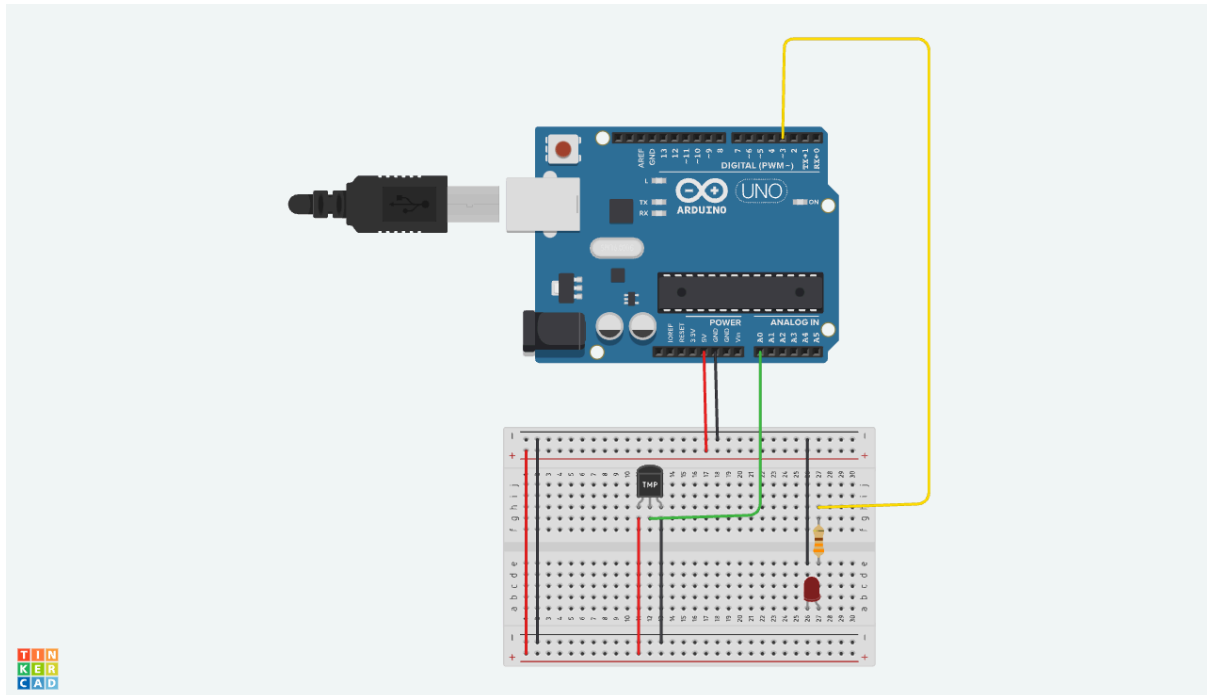
The program continuously compares the temperature value with a predefined threshold value of 50°C. If the measured temperature exceeds this limit, the Arduino activates the LED connected to digital pin 3 and displays a warning message "High Temperature!!" on the Serial Monitor.

If the temperature is below the threshold value, the LED remains turned OFF.

This process runs continuously in a loop, allowing the system to monitor environmental temperature in real time.

The entire process repeats continuously with a delay of one second between each reading for accuracy of reading.

CIRCUIT DIAGRAM:



Circuit diagram of smart temperature monitoring system

ARDUINO CODE:

```
// SMART TEMPERATURE MONITOR

const int sensorPin = A0; // defining input analog pin

const int led_Pin = 3; // defining output digital pin

void setup() {
  // begin serial monitor
  Serial.begin(9600);

  // declare output pin
  pinMode(led_Pin, OUTPUT);
}
```

```
void loop() {  
  
    // read ADC value from TMP36  
  
    int sensorValue = analogRead(sensorPin);  
  
  
    // Convert ADC value to voltage  
  
    float voltage = sensorValue * (5.0 / 1023.0);  
  
  
    // TMP36 temperature calculation  
  
    float temperatureC = (voltage - 0.5) * 100.0;  
  
  
    // conversion of temperature  
  
    float temperatureF = (temperatureC * 9.0 / 5.0) + 32.0;  
  
  
    // print the temperature values in serial monitor  
  
    Serial.print("Temperature: ");  
  
    Serial.print(temperatureC);  
  
    Serial.print(" C | ");  
  
    Serial.print(temperatureF);  
  
    Serial.println(" F");  
  
  
    // set threshold value as 50.0, if temperature crosses threshold value led glows up  
    // and print warning in serial monitor  
  
    if (temperatureC > 50.0) {  
        digitalWrite(led_Pin, HIGH);  
    }  
}
```

```
    Serial.println("High Temperature!!");  
}  
  
else {  
    digitalWrite(led_Pin, LOW);  
}  
  
// 1 sec delay for accurate measurement  
  
delay(1000);  
}
```

Arduino Program Explanation:

The program starts by defining the sensor input pin and LED output pin. Inside the setup function, the Serial Monitor is initialized with a baud rate of 9600 bits per second, and the LED pin is configured as an output.

Inside the loop function, the Arduino continuously reads the analog value from the TMP36 sensor using the `analogRead()` function.

The analog value is converted into voltage using the ADC conversion formula. The voltage is then converted into temperature in Celsius and Fahrenheit using mathematical calculations.

The temperature readings are displayed on the Serial Monitor for monitoring purposes.

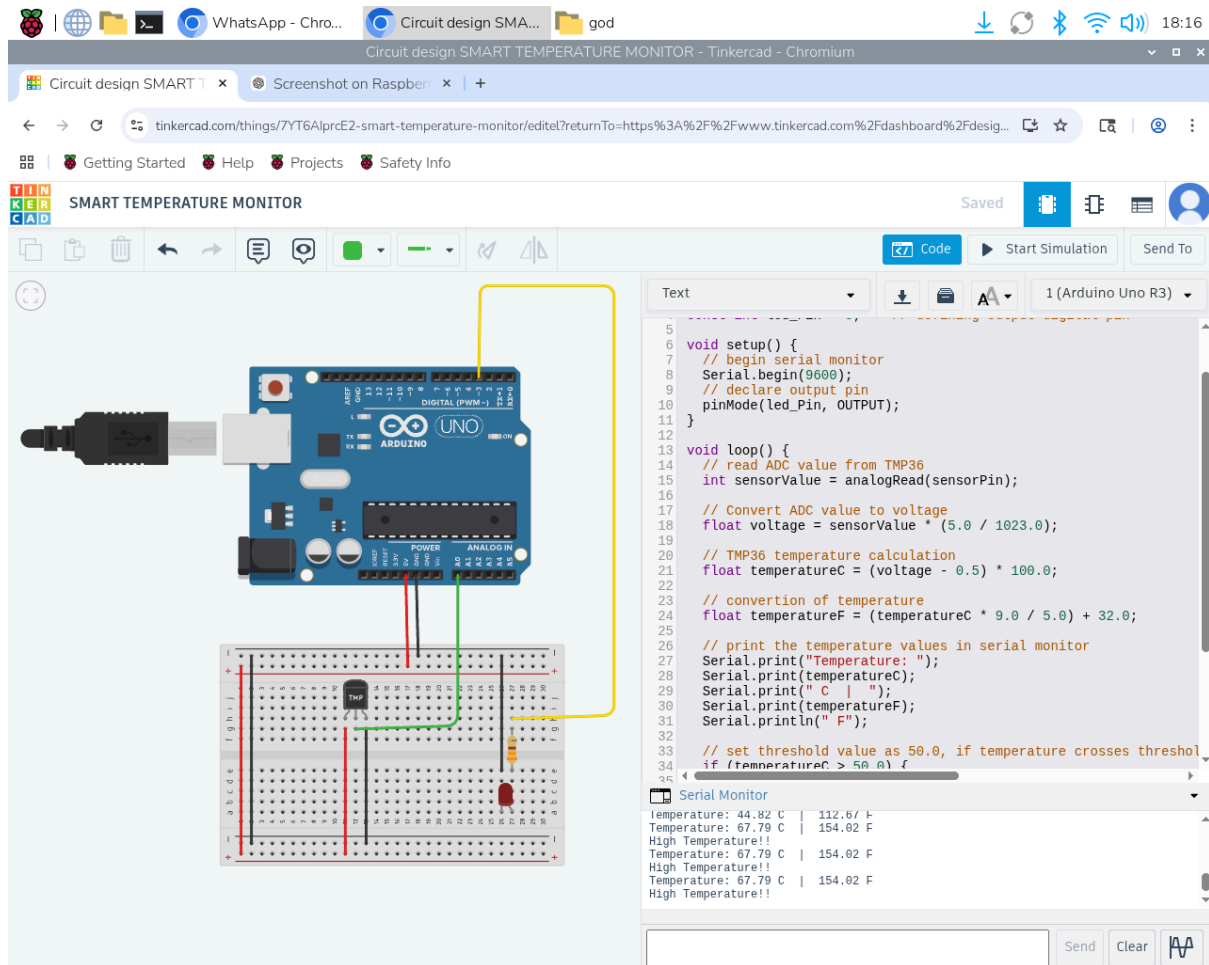
The program checks whether the temperature exceeds 50°C. If the condition becomes true, the LED connected to digital pin 3 turns ON and a warning message is printed. Otherwise, the LED remains OFF.

A delay of one second is added to ensure stable and accurate readings.

Output:

The system continuously displays temperature values in both Celsius and Fahrenheit on the Serial Monitor. When the temperature exceeds 50°C:

- LED glows
- Warning message appears



Simulation results and serial monitor output

Example Output:

Temperature: 29.45 C | 85.01 F

Temperature: 30.12 C | 86.21 F

Temperature: 51.30 C | 124.34 F

High Temperature!!

When the temperature exceeds 50°C , LED glows and a warning message “High Temperature!!” appears in the serial monitor.

Real-World Applications of the Smart Temperature Monitoring System:

Industrial Equipment Monitoring

The system can be used in industries to continuously monitor the temperature of machines, motors, and electrical equipment. If overheating occurs, the system provides an alert, helping prevent equipment damage and improving operational safety.

Smart Home Automation

In smart homes, the project can monitor room temperature and automatically control cooling devices such as fans or air conditioners. This improves comfort while also reducing unnecessary power consumption.

Healthcare and Medical Storage

Hospitals and medical laboratories require proper temperature control for storing medicines, vaccines, and medical equipment. This system helps maintain safe storage conditions by providing alerts when temperature exceeds the required limit.

Server Room Monitoring

Computer servers and networking devices generate heat during operation. The temperature monitoring system helps maintain safe environmental conditions in server rooms and prevents hardware failure caused by overheating.

Agriculture and Greenhouse Systems

The project can be used in greenhouses to monitor environmental temperature for healthy plant growth. Farmers can maintain suitable conditions for crops and improve agricultural productivity.

Conclusion:

The Smart Temperature Monitoring System successfully demonstrates the working of embedded systems using Arduino UNO and TMP36 temperature sensor. The project continuously measures environmental temperature, processes the sensor data, and generates alerts whenever the temperature exceeds a predefined threshold value.

This project provides a good understanding of sensor interfacing, analog-to-digital conversion, real-time monitoring, and automation concepts used in embedded systems and IoT applications.